

Department of Computer Engineering
National Institute of Technology, Kurukshetra-136119
Mid Term- I, Odd Semester 2025-2026

Subject: Design and Analysis of Algorithm (CSPC 201/DSPC 203)

Time: 50 Minutes

Maximum Marks: 15

Branch: B.Tech. M&C, AI-Data Science (III Semester)

Instructions: All questions are compulsory.

1	(a) Solve the recurrence relation: $T(n) = 2T(\sqrt{n}) + \lg n$ (b) Find the asymptotic bound for the sum $S(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$	2 2
2	Write pseudocode to compute the value of x^n (where x is the base and n is a non-negative integer exponent) using the divide and conquer strategy. Write recurrence relation for the algorithm and analyze its time complexity.	4
3	A hash table of size $m = 13$ is implemented using open addressing with double hashing. the primary and secondary hash functions are $h_1(k) = k \bmod 13$, $h_2(k) = 1 + (k \bmod 11)$. You are required to insert the following sequence of keys into the table in the given order: $k = 18, 41, 22, 44, 59, 32, 31, 73, 85, 26$. (i) write the final contents of the hash table after all insertions. (ii) count total collision.	3
4	Write an algorithm to construct a max-heap from an array containing n elements with a time complexity of $O(n)$. Prove the time complexity of your algorithm.	4